SECURITY ANALYSIS (BBAT35F97) · TAPMI · BBA(H) SEMESTER V

AI-AUGMENTED SECURITY ANALYSIS

Valuing Hindustan Unilever, Sun Pharma and ICICI Bank, and building a long-term portfolio from them

FMCG
31% OF P

Hindustan Unilever

β 0.60 · CAGR 9.6%

PHARMA
19% OF P

Sun Pharma

β 0.65 · CAGR 8.7%

PRIVATE BANK
50% OF P

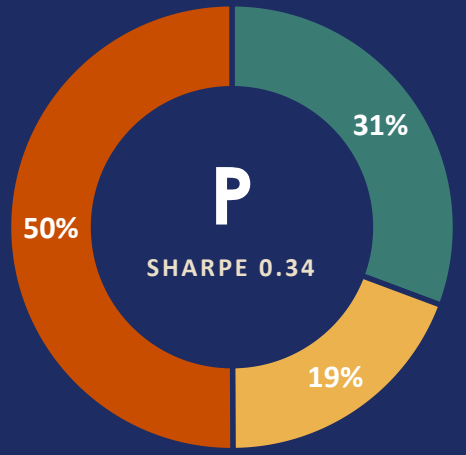
ICICI Bank

β 1.29 · CAGR 19.1%

DATA AS OF 13 OCT 2026 · STREET UPDATE 3 NOV 2026

HUL · SUN PHARMA · ICICI BANK

THE PORTFOLIO WE BUILT



₹64.8 lakh
MEDIAN 15-YEAR CORPUS FROM A ₹10,000 MONTHLY SIP

44 / 47
AI-EXTRACTED FIGURES WITHIN 1% OF YAHOO AND SCREENER

0 of 8
TRADING RULES THAT BEAT BUY-AND-HOLD AFTER COSTS

THREE SECTORS, THREE DIFFERENT DRIVERS

Four tests: a different macro driver, correlation of 0.5 or less, Nifty 50 liquidity with six quarters of transcripts, and a model that fits.

7.8%	+11.9%	5.25%	5.52%
REAL GDP GROWTH, Q1 FY27	INVESTMENT (GFCF) GROWTH	RBI REPO RATE	91-DAY T-BILL RATE

FMCG

Hindustan Unilever

DEFENSIVE ANCHOR

Lowest beta in the weakest sector: out-of-favour quality.

β 0.60 · CAGR 9.6%

PHARMA

Sun Pharma

DEFENSIVE GROWTH

Demand ignores the cycle; US sales earn dollars.

β 0.65 · CAGR 8.7%

PRIVATE BANK

ICICI Bank

GROWTH ENGINE

The direct play on investment-led growth: loans +19.6%.

β 1.29 · CAGR 19.1%

Sources: MOSPI, RBI, Zerodha Aftermarket Report (12 Oct 2026). Full funnel: docs/macro_industry.md.

WHY THEY BELONG TOGETHER

0.16–0.24

PAIRWISE CORRELATIONS, 2016–2026

Betas of 0.60, 0.65, 1.29: two defensives and one cyclical.

WHAT WE REJECTED

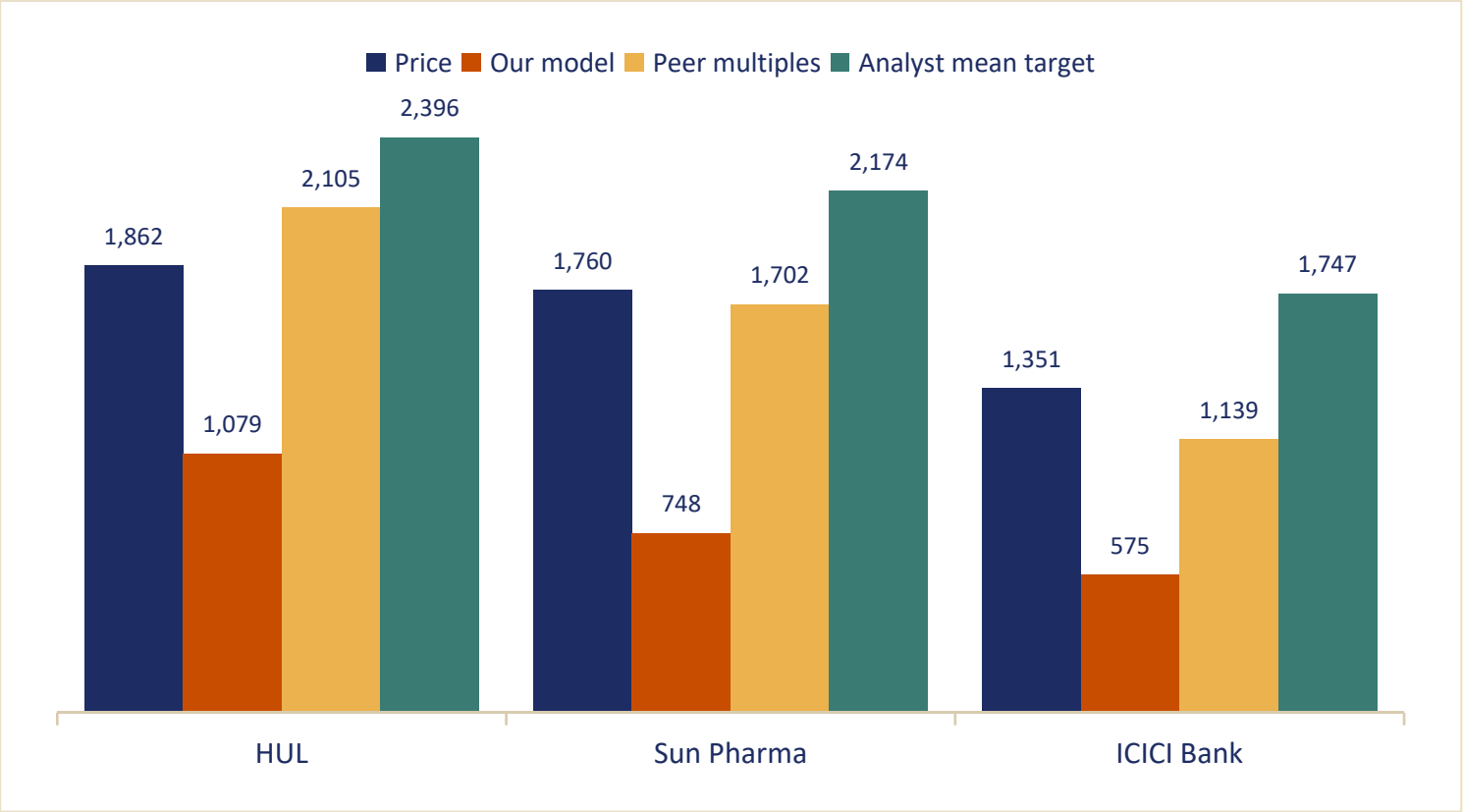
MEDIA
Strongest sector (+11.2% vs 200-day), but its certainty equivalent at A = 4 is −8% to −38%: chasing it is recency bias.

INFOSYS
IT is 8.9% below its 200-day; Sun keeps the dollar hedge at a lower beta.

HDFC BANK
The 2023 merger breaks its history.

WHAT THESE PRICES REQUIRE YOU TO BELIEVE

Intrinsic value from our Excel models at CAPM discount rates, against the 13 Oct price and the analysts' targets.



−14%

BLENDED VS PRICE

Hindustan Unilever

Price implies 7.4% growth a year for ever, at r = 9.8%

−30%

BLENDED VS PRICE

Sun Pharma

Price implies 8.6% growth a year for ever, at r = 10.1%

−37%

BLENDED VS PRICE

ICICI Bank

Price implies 13.9% growth a year for ever, at r = 14.7%

ICICI grew loans 19.6% last quarter. The gap is mostly our 14.7% cost of equity (β 1.29), not unrealistic growth.

Two-stage DDM (HUL), two-stage residual income (Sun Pharma, ICICI Bank), peer P/E and ROE-adjusted P/B: workbook/valuation.xlsx. Blended = average of model and peer values. Analyst targets: Yahoo Finance consensus.

THE STREET RANKS THEM ALMOST OPPOSITE TO OUR MODELS

Consensus on 3 Nov 2026, after the Q2 results. Our 13 Oct model inputs do not change.

Stock	Buy share	Mean target	Our r	r the price needs	r the target needs
HUL	69% of 39	₹2,256 (+17%)	9.8%	7.9%	7.6%
Sun Pharma	85% of 33	₹2,136 (+22%)	10.1%	7.6%	7.2%
ICICI Bank	100% of 42	₹1,769 (+26%)	14.7%	9.9%	9.1%

WHERE THE GAP COMES FROM

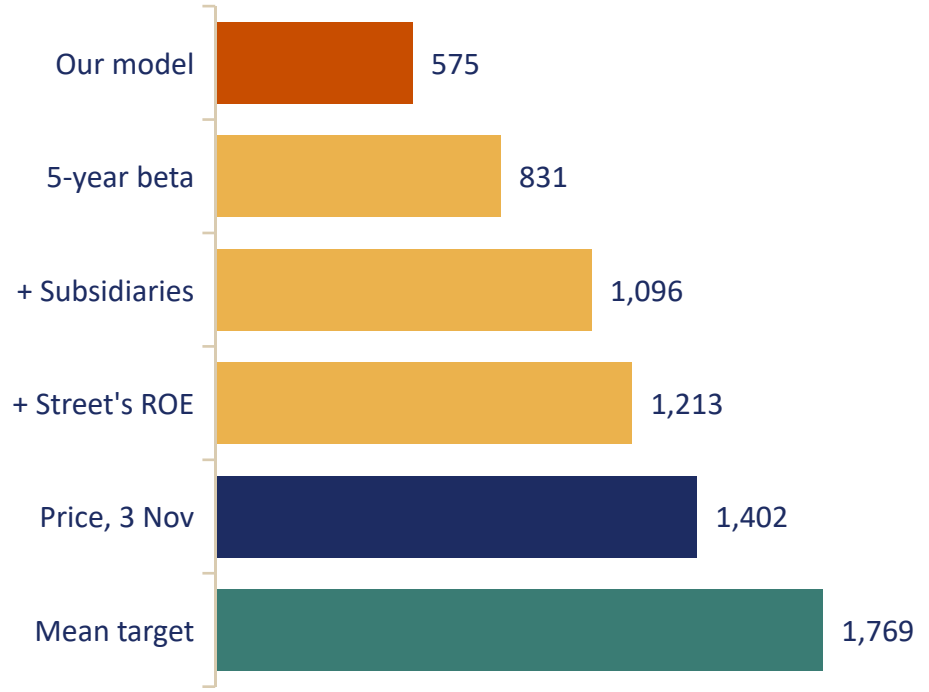
BETA WINDOW ICICI's beta is 1.29 over 10.8 years, 0.96 over five.

SUBSIDIARIES Brokers add about ₹265 a share; our residual-income model does not.

ROE The Street assumes 17% then 16%; we use 16% then 15%.

A difference in inputs, not facts. We keep our base case: changing beta after seeing the Street's answer would be tuning. With the 5-year beta, P would be 39 / 21 / 40.

ICICI BANK: FROM OUR VALUE TO THE PRICE, ONE ASSUMPTION AT A TIME



Yahoo Finance consensus, read 3 Nov 2026. Broker figures are mostly second-hand press summaries: data/street/street_views.csv marks each one.

THE MOSAIC: PUBLIC PIECES THAT TEST EACH THESIS

Mosaic theory allows public and non-material non-public information; ours is all public. Under semi-strong efficiency it tests a thesis; it does not promise alpha.

Hindustan Unilever

STREET
69% Buy; mean target cut 5.8% since 13 Oct. Morgan Stanley is Equal-weight.

Q2 FY27
Underlying volume +7%; margin down 142 bp to 22.6%.

WHO IS BUYING
FIIs 10.2% → 9.3%, DIIs 16.0% → 17.1% (Jun 25 to Sep 26).

WHAT IT CHANGES FOR US
Still the defensive anchor, but no bargain: "fairly valued" was a close call at the freeze and the price has since risen.

Sun Pharma

STREET
85% Buy; Bank of America says Underperform at ₹1,850.

Q2 FY27
India +17.3%; US -8.2% on lenalidomide erosion; reported profit −36% on Organon and litigation one-offs.

WHO IS BUYING
FIIs 17.3% → 14.4%, DIIs 19.4% → 22.3% (Jun 25 to Sep 26).

WHAT IT CHANGES FOR US
Organon ($11.75bn, all cash) means about 2.3× net debt/EBITDA. Our beta understates that, and the "defensive" label is weaker.

ICICI Bank

STREET
100% Buy; targets rest on 2.2–2.6× book value plus subsidiaries.

Q2 FY27
PAT +20%; NIM 4.18%, down 18 bp; Gross NPA 1.20%.

WHO IS BUYING
Not used: a jump in Mar 2026 looks like a reclassification.

WHAT IT CHANGES FOR US
Growth engine confirmed. The most liked stock is also 57% of our risk, so we watch crowding.

Yahoo Finance, Screener, press summaries of broker notes, read 3 Nov 2026. Sun Pharma's Q2 figures differ between sources.

FIVE AI TASKS, EACH CHECKED AGAINST A BENCHMARK

AI reads and labels; Python computes every number. Each task reports how far it can be trusted.

01 RATIO EXTRACTION

WHAT
Claude reads the FY26 annual reports and returns line items as JSON.

CHECKED AGAINST
Yahoo Finance and Screener.

44 / 47

within 1%; 0 extraction errors

02 SENTIMENT

WHAT
FinBERT scores 2,428 headlines and 9,133 call sentences.

CHECKED AGAINST
Management vs analysts on every call.

+0.28

HUL tone gap: management far more upbeat

03 BIAS & FRAMING

WHAT
An LLM tags 120 headlines for framing, authority, herding and hype.

CHECKED AGAINST
70% agreement with FinBERT.

41%

carry a bias cue; 14% aren't about the company

04 FORECASTING

WHAT
Moving-average and RSI rules, an earnings event study, analyst consensus.

CHECKED AGAINST
Buy-and-hold, after a 0.25% trading cost.

0 of 8

trading rules beat buy-and-hold

05 RISK HEATMAPS

WHAT
Correlation, rolling correlation, risk contribution, stress tests.

CHECKED AGAINST
Four real sell-offs since 2020.

57%

of portfolio risk is ICICI, at 50% weight

Prompts: docs/ai_prompts.md. Extraction: data/ai/ratio_extraction.json (page references for every figure). FinBERT: ProsusAI/finbert. Headlines: Google News RSS, last 12 months.

SEE IT LIVE

Two minutes, four stops on our dashboard.

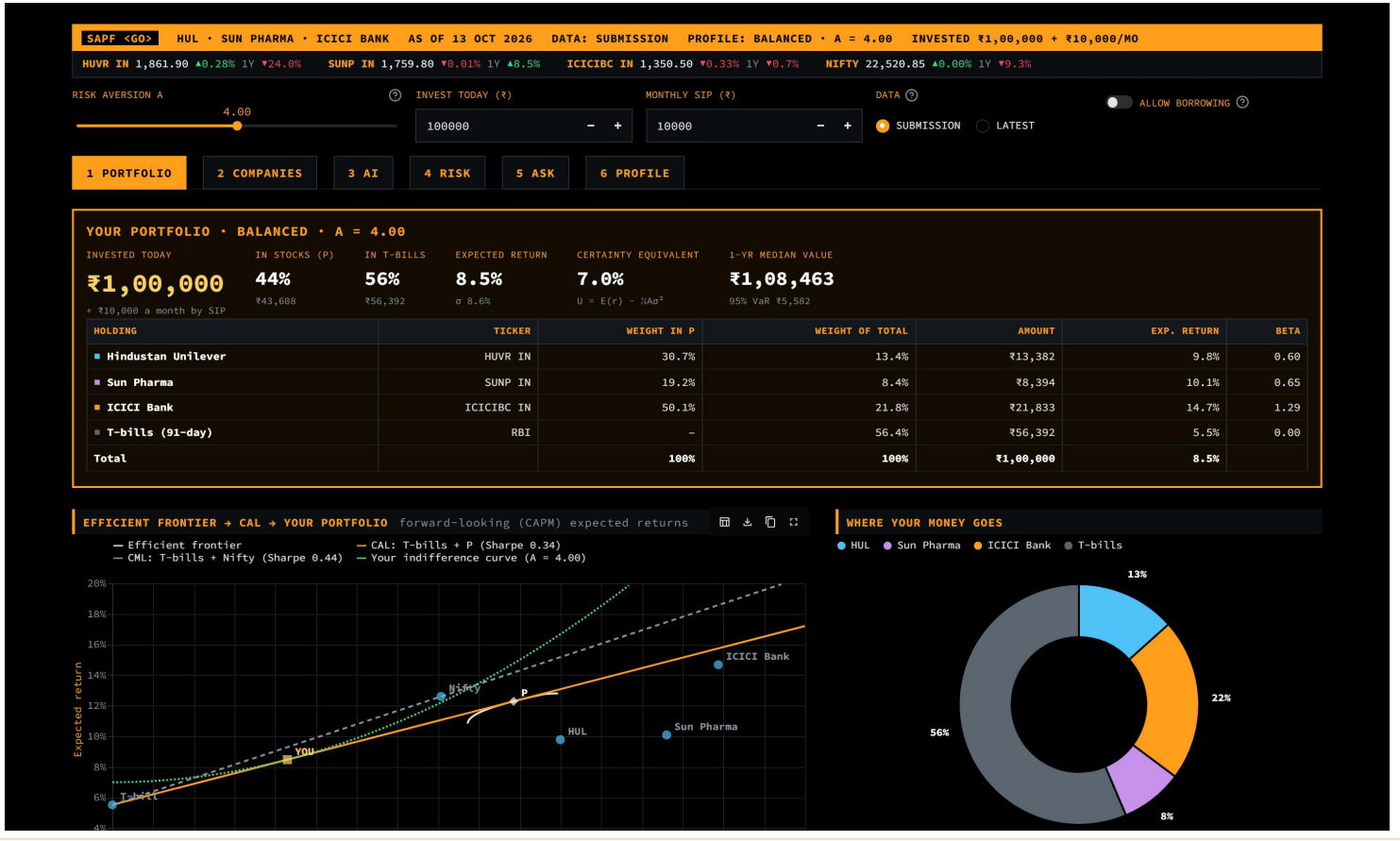
01 RISK QUESTIONNAIRE
Ten answers set your risk aversion A

02 YOUR PORTFOLIO
The rupee split for your A; slide A from 4 to 2: 44% → 87% in P

03 RISK
Your rupee loss in four real sell-offs

04 ASK THE PORTFOLIO
One question, answered with its sources

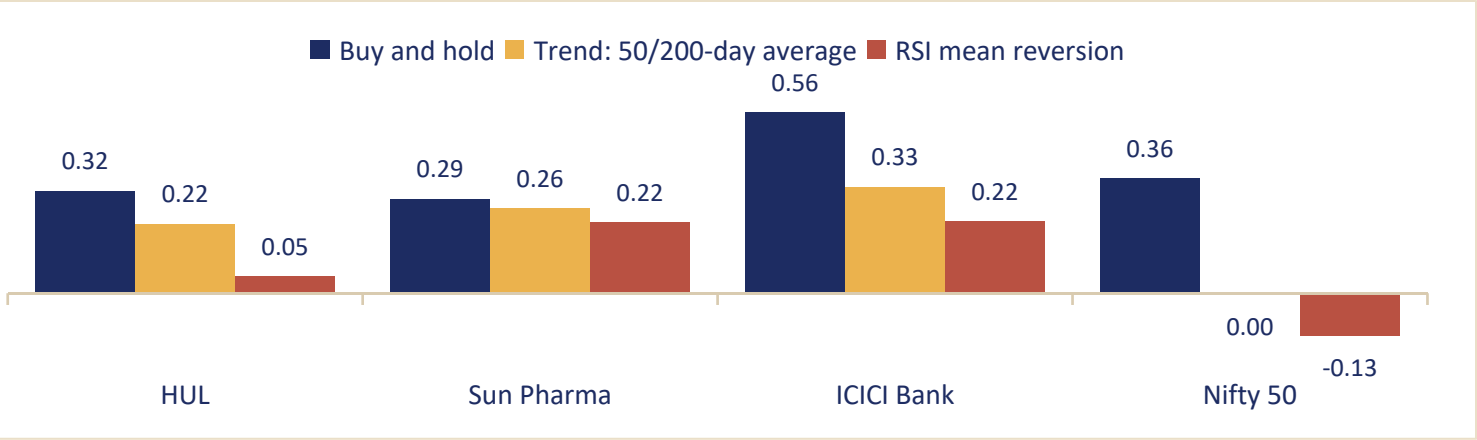
LOCAL COPY
streamlit run app.py



IF ANYTHING TAKES MORE THAN 10 SECONDS
1 Live link → 2 Local copy → 3 Recorded walkthrough → 4 Screenshots (A3)

WE FOUND NO RELIABLE WAY TO BEAT THESE PRICES

Top: Sharpe ratio of trading rules against buy-and-hold, after a 0.25% cost. Below: abnormal returns around 72 earnings results.



Window (trading days)	Beats (50)	Misses (22)	Beat minus miss [95% CI]
Before: −20 to −2	+1.16% (t 1.7)	+0.59% (t 0.5)	–
Reaction: −1 to +1	**+1.22% (t 2.2)**	**−1.57% (t −2.2)**	**+2.79% [+1.04%, +4.54%]**
After: +2 to +20	−1.47% (t −2.5)	−0.94% (t −0.9)	−0.53% [−2.80%, +1.74%]
After: +2 to +60	+0.52% (t 0.5)	−1.12% (t −0.7)	+1.64% [−2.21%, +5.49%]

Yahoo Finance, 2016–2026. Market-model abnormal returns; 250-day estimation window. Results: data/submission/emh.json.

WEAK FORM

0 OF 8

RULES THAT BEAT BUY-AND-HOLD

Autocorrelation and variance ratios: largest |z| is 1.9, below the 2 that would flag a pattern.

SEMI-STRONG FORM

+2.79%

BEAT MINUS MISS, IN THREE DAYS

After that, no drift we can detect: both later ranges include zero. But 72 events can miss a small drift; Harshita et al. (2018) found about 6% over 64 days in the Nifty 500.

STRONG FORM

Not tested: it needs inside information we do not have.

FEAR AND GREED DESCRIBES THE CROWD; IT DOES NOT TIME IT

CNN's Fear & Greed index is American. In India: Tickertape's Market Mood Index, India VIX, and our own score for each stock.

62.7

MARKET MOOD INDEX: GREED
3 Nov, one hand reading

16.2

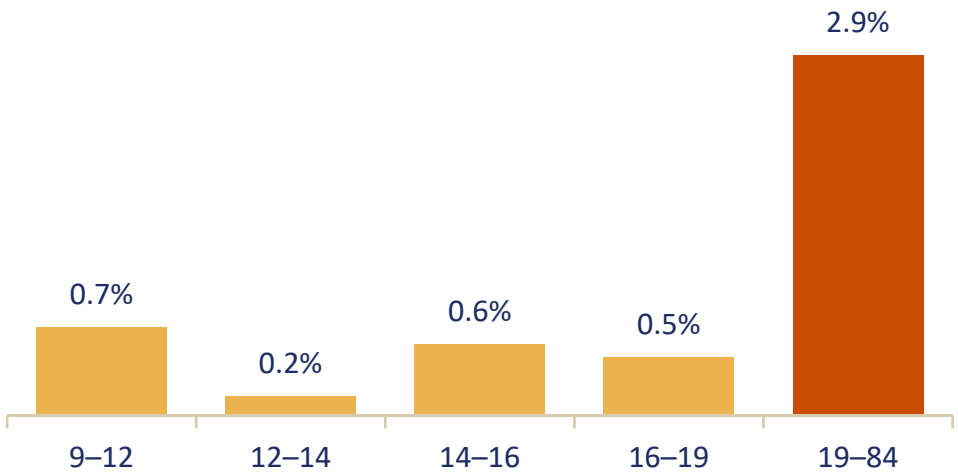
INDIA VIX
75% percentile of the past year

OUR STOCK SCORE, 0 = FEAR, 100 = GREED: 3 NOV (13 OCT)

39 FEAR	**37** FEAR	**58** GREED
HUL (17)	**Sun Pharma (18)**	**ICICI Bank (36)**

No stock-level test reaches p < 0.05. The score is built from prices, so it describes mood rather than measuring it independently. FinBERT tone says nothing about the next five days (|r| ≤ 0.08).

NIFTY OVER THE NEXT 20 DAYS, BY INDIA VIX LEVEL

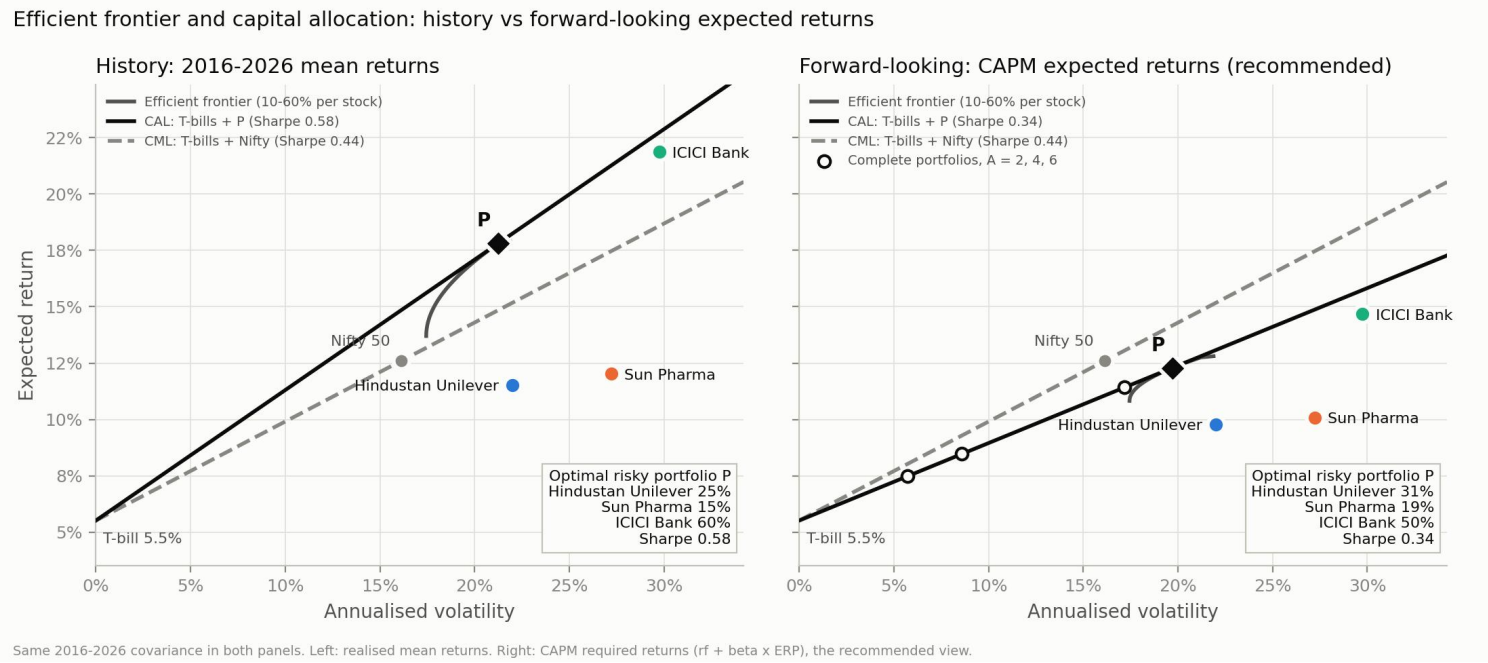


Fear has been followed by higher returns (r = 0.24, p = 0.005; without COVID, r = 0.20, p = 0.024), but one quintile carries it, and it may be a risk premium rather than a mispricing.

Tickertape MMI, 3 Nov 2026, 9:30 AM. ^INDIAVIX, Yahoo Finance. Score: momentum, RSI, calm and distance from the high, ranked over 750 days.

31 / 19 / 50, SIZED TO EACH INVESTOR

Forward-looking (CAPM) expected returns, 10–60% per stock. History alone would put 60% in last decade's winner.



87%
IN P

AGGRESSIVE · A = 2
E(r) 11.4% · σ 17.2%
Certainty equivalent 8.5%

44%
IN P

BALANCED · A = 4
E(r) 8.5% · σ 8.6%
Certainty equivalent 7.0%

29%
IN P

CONSERVATIVE · A = 6
E(r) 7.5% · σ 5.7%
Certainty equivalent 6.5%

SHARPE 0.34 vs 0.44

Three stocks don't beat the index on risk-adjusted terms, so the SIP pairs them with a Nifty core. Out of sample (2023–26): forward P +7.9%/yr, Nifty +7.4%, history-based P +3.2%.

DID IT BEAT THE MARKET? A LITTLE, BUT NO ALPHA WE CAN PROVE

Realised returns. 2023–26 weights were fitted on 2016–22, so that block is a real test; the 2016–26 block is the stocks over the whole sample.

Portfolio	CAGR	Rp − Rf	Rp − Rm	σ	β	Sharpe	Treynor	Jensen's α	t(α)
2023–26: Forward-looking P	**7.9%**	**+3.0%**	**+0.5%**	**12.8%**	**0.71**	**0.24**	**4.3%**	**+1.2%**	**0.27**
2023–26: Equal weight	8.7%	+3.8%	+1.2%	12.3%	0.65	0.31	5.8%	+2.1%	0.47
2023–26: History-based P	3.2%	−1.3%	−3.8%	13.7%	0.65	−0.09	−2.0%	−3.0%	−0.55
2023–26: Nifty 50	7.4%	+2.5%	–	12.6%	1.00	0.20	2.5%	+0.0%	–
2016–26: HUL	9.6%	+6.2%	−1.1%	22.0%	0.60	0.28	10.2%	+1.8%	0.32
2016–26: Sun Pharma	8.7%	+6.7%	−0.6%	27.2%	0.65	0.24	10.3%	+2.0%	0.26
2016–26: ICICI Bank	19.1%	+16.5%	+9.3%	29.7%	1.29	0.55	12.8%	+7.1%	1.17
2016–26: Nifty 50	12.0%	+7.2%	–	16.1%	1.00	0.45	7.2%	+0.0%	–

SHARPE = (Rp − Rf) / σ

0.24 vs 0.20

Rewards total risk: the test for a whole portfolio.

TREYNOR = (Rp − Rf) / β

4.3% vs 2.5%

Rewards market risk: the test for a piece of a diversified one, like our satellite.

ALPHA = (Rp − Rf) − β(Rm − Rf)

+1.2%, t 0.27

No α here reaches t = 2. On your money it scales with y: 0.53% at A = 4.

Newey-West t-statistics, 5 lags. Nifty 50 with estimated dividends. Rp − Rm is the plain gap to the index. Results: data/submission/performance.json.

OVER THE YEARS

₹10,000 a month for 15 years, under our Investment Policy Statement.

RETURN
Beat 4% inflation clearly; target ₹75 lakh

RISK
Balanced (A = 4) unless the questionnaire says otherwise

LIQUIDITY
6 months of expenses kept outside the portfolio

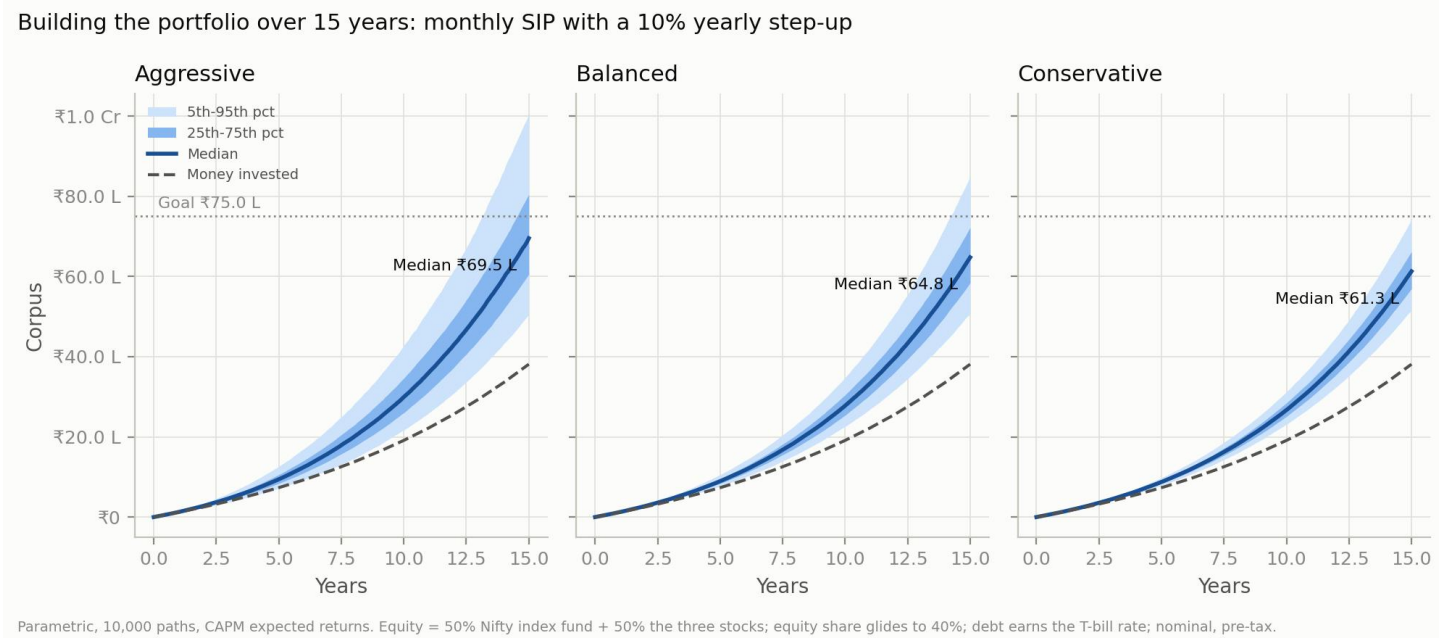
REBALANCE
Once a year, with new SIP money first

BEHAVIOUR
No selling in the month after a 20% fall

₹38.1 lakh
INVESTED (10% STEP-UP A YEAR)

₹64.8 lakh
MEDIAN CORPUS, BALANCED

19% / 11%
CHANCE OF ₹75 LAKH (TWO MODELS)



50% Nifty 50 index fund + 50% our three stocks, gliding to 40% equity. Parametric and bootstrap Monte Carlo, 10,000 paths, pre-tax. Aggressive: 36% / 29% chance of ₹75 lakh.

OUR CALL

Range: our blended value to the analysts' mean target. Prices of 13 Oct.

HINDUSTAN UNILEVER

HOLD

₹1,592–2,396 · price ₹1,862

SUN PHARMA

HOLD

₹1,225–2,174 · price ₹1,760

ICICI BANK

HOLD

₹857–1,747 · price ₹1,351

HOLD AT THESE WEIGHTS, ADD ONLY THROUGH THE SIP

Good businesses at full prices: own them, but let monthly buying average the entry price.

RISKS WE ARE WATCHING

01 **CONCENTRATION** ICICI is 50% of the weight but 57% of the risk.

02 **VALUATION** Prices require 7.4% to 13.9% growth a year for ever, or a lower discount rate.

03 **SUN PHARMA** Organon's debt, and proposed caps on trade margins for anti-cancer drugs.

LIMITATIONS

One risk-free rate for 2016–26 · Nifty total return estimated (+1.22%/yr dividends) · Sun Pharma chosen after its run · broker views mostly second-hand press summaries · pre-tax, no fees · coursework, not investment advice.

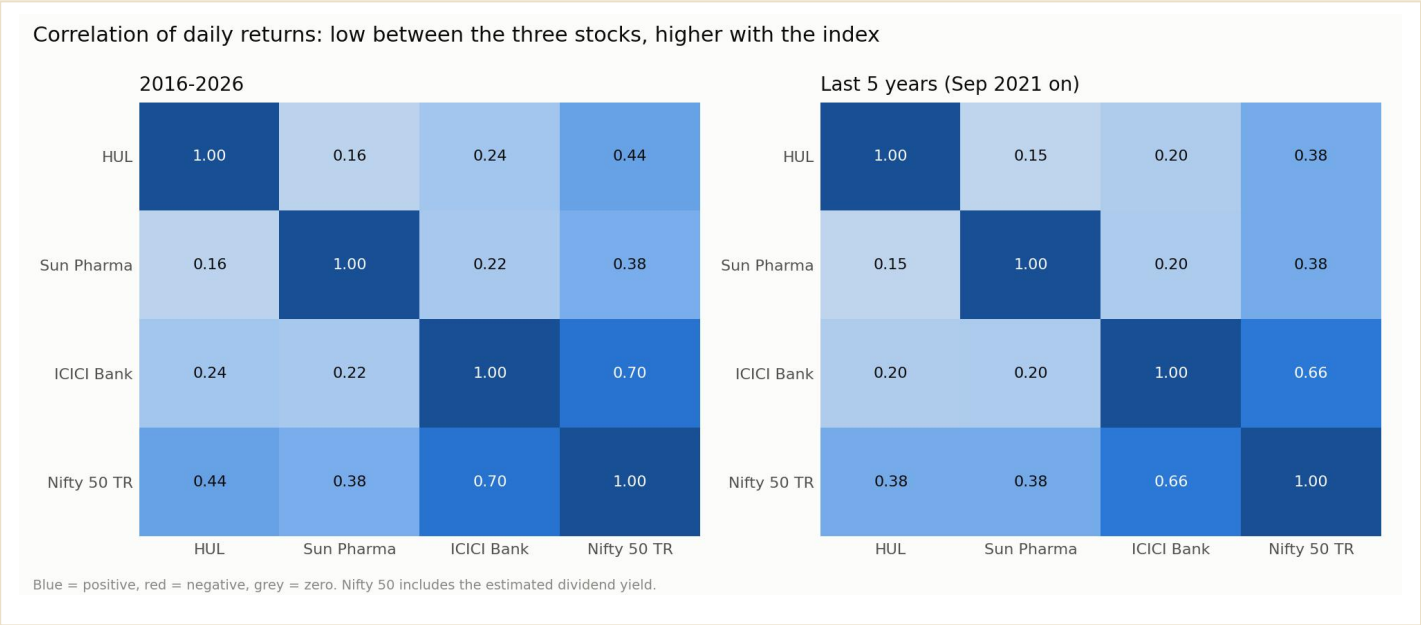
METHODS AND ASSUMPTIONS

Every number in this deck is computed from these choices; the code is in our GitHub repository.

Item	Choice	Source
Prices	Daily adjusted closes from 2016; HUL demerger (Dec 2025) adjusted	Yahoo Finance; NSE special pre-open
Risk-free rate	5.52%, 91-day T-bill	RBI, 1 Oct 2026
Equity risk premium	7.08% (India)	Damodaran, 5 Jan 2026
Benchmark	Nifty 50 price index + 1.22%/yr estimated dividends	Nifty BeES vs index gap, 2016–2026
Valuation	Two-stage DDM (HUL); two-stage residual income (Sun Pharma, ICICI); peer multiples; r × g grids	workbook/valuation.xlsx
Optimisation	Maximum Sharpe ratio, 10–60% per stock, CAPM expected returns (rf + β × ERP)	scipy SLSQP
Complete portfolio	$y^* = (E[rP] - rf) / (A\sigma^2)$, capped at 100%; A = 2, 4, 6	BKM Ch 6
Monte Carlo	10,000 paths, 1 year; normal (Cholesky), 10-day block bootstrap, EWMA λ = 0.94	numpy, seed 42
SIP	₹10,000/month, +10%/yr, 15 years; 50% Nifty core; glide to 40% equity	Pre-tax, no fees
AI	FinBERT sentiment; Claude for extraction, aspects and headline tagging	docs/ai_prompts.md
Efficiency tests	Robust lag-1 autocorrelation, Lo-MacKinlay variance ratios, event windows to day +60	src/emh_tests.py

WHERE THE RISK SITS

Correlations of daily returns, and what happened to the portfolio in four real sell-offs.



Episode	Our P	Nifty 50
COVID crash (Feb-Mar 2020)	**−34.6%**	−37.2%
Rate-hike sell-off (Jan-Jun 2022)	**−12.3%**	−16.1%
Election-result shock (4 Jun 2024)	**−2.3%**	−5.9%
2026 drawdown (02 Jan 2026 to 13 Oct 2026)	**−5.1%**	−13.7%

Our portfolio fell less than the Nifty in 4 of 4 episodes.

32%

HUL
SHARE OF PORTFOLIO RISK

11%

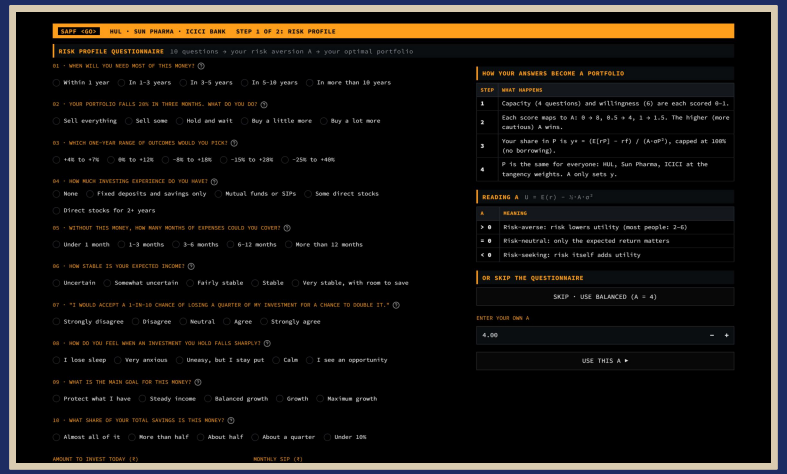
SUN PHARMA
SHARE OF PORTFOLIO RISK

57%

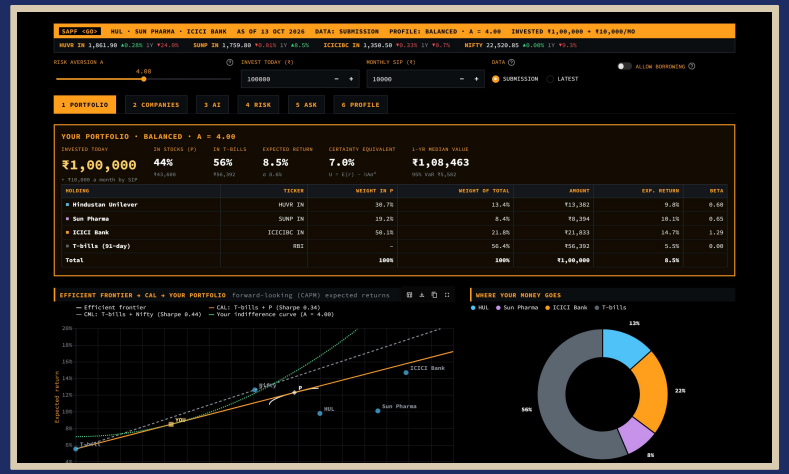
ICICI BANK
SHARE OF PORTFOLIO RISK

THE DASHBOARD, IF THE DEMO FAILS

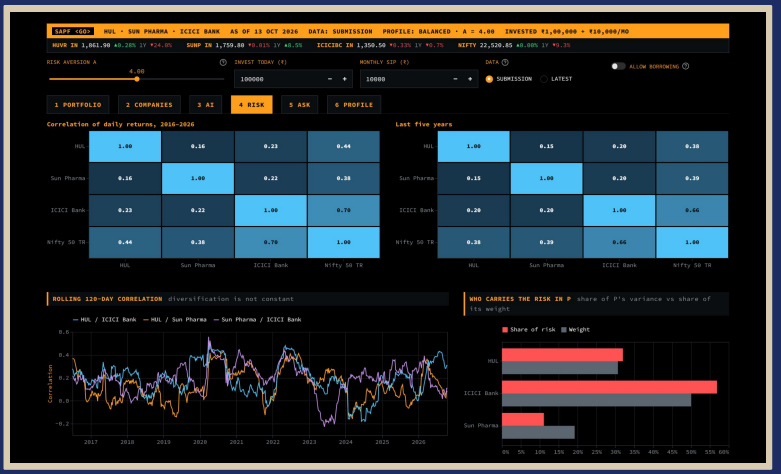
Rung 4 of the backup ladder: the same screens as the live demo.



STEP 1: RISK QUESTIONNAIRE



YOUR PORTFOLIO IN RUPEES



RISK: HEATMAPS AND STRESS

DATA SOURCES AND AI PROMPTS

Free and public sources only; every AI prompt is saved with its output.

DATA

- Prices, statements, analyst consensus: Yahoo Finance (yfinance)
- Ten-year financials: Screener.in public pages
- Annual reports and earnings-call transcripts: BSE filings and company IR sites
- Risk-free rate and repo rate: Reserve Bank of India
- GDP and demand mix: MOSPI press note, 31 Aug 2026
- Equity risk premium: Damodaran, India, Jan 2026
- Headlines: Google News RSS, 12 months
- Sector momentum: Zerodha Aftermarket Report, 12 Oct 2026
- Broker views: mostly press summaries, marked in data/street/street_views.csv
- Mood: Tickertape MMI (one reading, 3 Nov 2026); India VIX

AI AND METHOD

- FinBERT: ProsusAI/finbert (Hugging Face)
- Claude (Anthropic): ratio extraction, concall aspects, headline tagging
- Prompts P1–P5 and outputs: docs/ai_prompts.md, data/ai/
- Bodie, Kane & Marcus, Investments: Ch 6, 7, 9, 11, 12, 17, 18, 24
- Harshita, Singh & Yadav (2018), post-earnings drift in the Nifty 500; SPIVA India
- Valuation workbook: workbook/valuation.xlsx (live formulas)
- Code, tests and the nightly refresh: GitHub repository